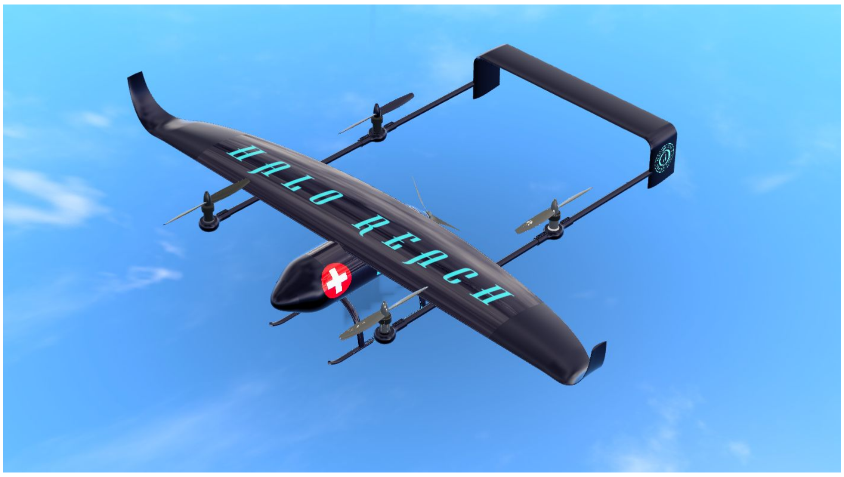
HALO REACH: Material, Manufacturing, Undercarriage, Boom and Dynamics

UAS for Disaster Relief - AE3 2024



Student: Zhiyan Wang
CID: 01911209

Supervisor: Dr. Siti Ros
Department: Department of Aeronautics

Course: MEng Aeronautical Engineering
Module: AERO60004 - Group Design Project
Date: 17/06/2024
Academic Year: 2023-2024

Department of Aeronautics
South Kensington Campus
Imperial College London

London SW7 2AZ
U.K.

Table 1. People involved in HALO REACH - GDP (**student in bold**)

Team code	**Team**	**Names**	CID
FATT01	Team Leads	Moreno, Patricia	01857449
		Rashid, Hasan H	02020313
FATT02	Aerodynamics	Alam, Shahriar	02037395
		Dirirsa, Naboth	02026211
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
		Allen, Harry	02028116
		Tang, Shun S	02023280
		Lee, Jeongheon (Edward)	01754474
FATT06	Add-ons	Rosenthal, Adam T	02027024
		Ramteke, Shreya	02079050
		Nakhla, Andrew	02044646

Abstract

With careful comparison the material for the aircraft was determined to be Toray Prepreg 3960-19, with different layups for different components.The undercarriage was designed using landing conditions from literature while considering the requirements of this specific case, verified using both static analysis with Ansys Mechanical and dynamic simulation through Ansys LS Dyna. The layup of the material is designed as two layers of $(90/45/-45/0/-45/45/90)_s$ layup according to the results of the FEA, which is a total thickness of 2.56mm. A convergence study was performed. The boom loading was then analysed. After that the dynamic performance of the entire aircraft was evaluated by first doing a simplified version of the aircraft and then a dynamic analysis using Ansys Mechanical. Manufacturing process of the aircraft components are then designed accordingly.

Contents

List of Figures

List of Tables

Nomenclature

$\ddot{h}$	Acceleration in y-direction (m/s^2)
m_{total}	Total mass (kg)
N	Normal force (N)
P	Power (W)
x_T	Thrust moment arm (m)
GTOW	Gross Takeoff Weight
HALO REACH	Highly Aerodynamic Light Operation Remote Emergency Aid & Crisis Help
LQR	Linear Quadratic Regulator
MI	Merit Index
MPC	Model Predictive Controller
PIDF	Proportional, Integral, Derivative, and Filter
RMS	Root Mean Square
UAS	Uncrewed Aerial System
VTOL	Vertical Take-Off and Landing

1 Introduction

In the context of this UAV, selecting the right material becomes critical not only for ensuring its operational efficiency but also for enhancing its durability and reliability in harsh conditions. The undercarriage, being a fundamental component, must be constructed from materials that can withstand significant mechanical stress while maintaining a lightweight profile to support agile maneuvers and extended flight times. Advanced composites and alloys, known for their exceptional strength-to-weight ratios, are potential candidates.
Moreover, the UAV must be designed to function effectively in diverse and unpredictable environments, ranging from high-temperature zones to areas with heavy debris or moisture. This calls for materials that are not only heat-resistant but also capable of withstanding corrosion and wear over time. The integration of such materials ensures that the UAV can operate continuously without frequent maintenance, thereby reducing downtime and operational costs.
From a manufacturing perspective, it is essential to employ materials that are not only robust and durable but also cost-effective and easy to work with in large-scale production. The use of advanced manufacturing techniques such as additive manufacturing and CNC machining can further optimize material usage, reduce waste, and improve the overall cost efficiency of the UAV production process. By carefully selecting and combining materials with these considerations in mind, the UAV can achieve a balance between performance, durability, and cost, making it a competitive and reliable solution for disaster relief operations.

2 Material Selection

The first step of material selection is to analyze the requirements for different components and find the appropriate material merit index by analysing the loading conditions and identifying what is the most important property of the material. In order for this UAV to be competitive in the market,the main concern is to have the UAV as lightweight and low cost as possible. with that in mind, merit indices with the priority of minimum mass is chosen. The specific equations are listed below.

$$\text{spars} = \frac{E}{\rho} \tag{1}$$

$$\text{ribs} = \frac{\sqrt{E}}{\rho} \tag{2}$$

$$\text{boom\_tail} = \frac{\sqrt{E}}{\rho} \tag{3}$$

$$\text{top\_wing\_skin} = \frac{\sigma_c}{\rho} \tag{4}$$

$$\text{bottom\_wing\_skin} = \frac{\sigma_t}{\rho} \tag{5}$$

$$\text{fuselage\_skin\_light} = \frac{\sigma_t}{\rho} \tag{6}$$

$$\text{fuselage\_skin\_heavy} = \frac{E_c^{\frac{1}{3}}}{\rho} \tag{7}$$

$$\text{rotor\_blade\_skin} = \frac{G}{\rho} \tag{8}$$

$$K_{IC} = \frac{\sigma_t\sqrt{E \cdot 10^9}}{10^6} \quad \text{(K\_IC in MPa}\cdot\text{m}^{0.5}\text{)} \tag{9}$$

to have cost taken into account, all he indices will be divided by the price of that material.
After the material merit indices were chosen, the materials were selected. Large companies were looked at due to the completeness of their data sheet, wide range of products and detailed manufacturing instructions. for this UAV, due to cost limited design,only prepregs were taken into consideration. Hexcel and Toray were two big companies that has (list advantages of choosing big companies) since most materials are already classified

by the companies based on their use, it is easy to identify some potential targets. the ones listed by the company as suitable for aircraft and UAV are all chosen along with some high performance material that has outstanding properties. at a closer look, due to calculation and manufacturing ease considerations only unidirectional tape materials were taken as potential choices. the materials in the final step of analysis is listed as below, where TS stands for Tensile strength, CS stands fore compressive strength and GSM stands for Grams per Square Meter:

Material	**Ply t (mm)**	E_t **(GPa)**	E_c **(GPa)**	G **(GPa)**	ρ **(kg/m³)**	**TS (MPa)**	**CS (MPa)**	**GSM**
Hexcel M56//IM7-12K	0.253	182	156	4.5	1530	2730	1550	394
Hexcel M56/AS7-12K	0.134	141	129	4.1	1530	2330	1308	206
Toray 2700	0.152	59	54.7	4.16	1566.22	1029	756	230.8
Toray 4000-T800G	0.147	157	119.09	5	1272	2845	1558	230.8
Toray 4000-T1100G	0.145	180	136.54	5.2	1566.7	3174	1397	223.1
Toray 2510	0.152	125	112	4.23	1559.1	2172	1448	230.8
Toray 3960-19L	0.191	148	130	3.94	1566.1	3006	1779	296.1
Toray 3900-145	0.218	69	63.6	3.6	1566	1034	668	224.8
Toray 3960-19	0.183	175	153	5.37	1562	3941	2048	288.7
Toray 3960-7	0.074	155	134	4.1	1524.5	3282	1387	116.7
Toray 3960-145	0.145	71	8.27	254	1553.3	3282	155	78.1
Toray TC275-1E	0.153	168	164	3.7	1505.3	2719	1418	230.8
Toray TC380	0.149	183	143	3.7	1522	2923	1496	227.3

Table 2. Material properties and measurements

the Grams per Square Meter data was not given directly by the data sheets but was obtained from the official calculator provided by each company. for some materials the ply thickness from the calculator differs from the data sheet, but the difference was minor and are all within 3%, thus can be considered reliable, and the final values used was the ones from the data sheets. With these data acquired, the material indices can be obtained. Given the constraints and merit equations,Toray 3960-19 stood out for all applications. The detailed indices table can be found in the appendix. However, the heavy frame in the fuselage structure used Al 2024-T4 [1]. Toray 3960-19 prepreg system is not only be best functioning choice out of all the materials considered, the matrix also has high heat resistance and is described by the company as high impact tolerance, indicating its ability to endure the harsh operational environments.

3 UnderCarriage Design

3.1 Configuration

when exploring the undercarriage design, a few configurations were taken into account, namely skids, wheels, legs and adaptive. the advantages and disadvantages of each design is listed as below.

Undercarriage Type	Advantages	Disadvantages
Skids	1. Lightweight 2. Simple and low maintenance 3. Effective on rough terrain	1. Limited ground mobility 2. Can be less stable on smooth surfaces 3. Increased landing impact
Wheels	1. Excellent ground mobility 2. Smooth takeoffs and landings 3. Suitable for paved surfaces	1. Heavier and more complex 2. Requires maintenance 3. Less effective on uneven terrain
Legs	1. Stable on uneven surfaces 2. Retractable to reduce drag 3. Good shock absorption	1. Mechanically complex 2. Increased weight 3. Requires more maintenance
Adaptive	1. Versatile for various terrains 2. Combines features of skids and wheels 3. Dynamic adjustment capabilities	1. Highly complex 2. Heaviest option 3. Expensive to develop and maintain

The simplicity of skids, with fewer moving parts and straightforward design, minimizes the risk of mechanical failure. Disaster relief scenarios often involve challenging environments where maintaining equipment can be difficult and time-consuming. The low maintenance requirement of skids means that the UAV can remain operational with minimal downtime, which is vital for continuous support in emergency situations. The simplicity of skids also allows for easier and faster repairs if needed, ensuring the UAV can be quickly redeployed. Disaster zones frequently feature rough, uneven, and debris-covered terrain where traditional wheeled undercarriage systems might struggle. Skids are specifically designed to handle such conditions with ease. They provide a broad, stable base for landing and takeoff, reducing the risk of the UAV tipping over or becoming stuck. This capability is essential for operations in environments where infrastructure is damaged or non-existent, allowing the UAV to land and take off from a variety of surfaces, including dirt, rubble, or uneven ground. With the above considerations in mind, the skids configuration was determined to be the best choice of the four.

3.2 Shape of Undercarriage

The shape of the undercarriage requires multiple angles of approach. Since the material is already chosen to be Toray 3960-19, which is a carbon fiber reinforce epoxy system, although it is very much less likely to be damaged compared to other materials such as thermoplastics or aluminium, any cracks or damage to the structure is not as easily repairable. thus having identical undercarriages on each side is crucial to the maintenance, meaning one spare skid part can be a backup for both the left and right skid, significantly reducing the pressure of logistics.

According to the landing conditions the aircraft should have an undercarriage capable of handling 7 degrees tilted landing. With the constraints given by the payload, which is a minimum of 40cm height from the bottom of the fuselage, the wing tips and tail plane will not touch the ground when landing and has a very safe clearance. However, for a better landing process that causes less stress in the skid, warped tips are employed so that during tilted landing the region touching the ground is relatively parallel to the ground instead of a simple tip.

For the ease of manufacturing a circular or oval shape is preferred over triangular or square cross sections. The oval-shaped cross-section of a skid offers several critical advantages that enhance the performance and reliability of disaster relief UAVs. Its smooth, continuous surface significantly reduces susceptibility to wear and tear compared to the sharp edges and corners of a square skid, which extends the skid's lifespan and lowers maintenance requirements. This durability is particularly valuable for UAVs operating in remote or difficult-to-access disaster areas, where maintenance opportunities are limited. Additionally, the oval shape provides a broader and flatter contact surface with the ground, improving stability during landings and takeoffs and reducing the risk of tipping over, especially on uneven or soft terrain. This shape also ensures a more effective distribution of the UAV's weight across the skid's contact area, thereby decreasing pressure points and minimizing the likelihood of the skid sinking into softer ground. Such characteristics are crucial in post-disaster environments where ground conditions may be unstable, allowing the UAV to maintain operational

effectiveness in challenging settings.
Arms also have oval cross sections, where the longer axis is along the airflow direction, leading to less drag, thus higher energy efficiency compared to a circular one while keeping the ease of maintenance and manufacturing. the arms were placed 80cm apert so that the moment arm of the tips are as short as possible and no attachment plate touches the contracted area
The length of the skid arms should be as short as possible. so that the undercarriage remains a light weight, since the payload requires 40cm of clearance and the undercarriage is attached to the sides of the fuselage, the total height of the undercarriage is determined to be 43cm. the width between the attachment point to the skid is 25cm, giving a proper width to attach and remove the payload and prevent tipping over during landing with a wider undercarriage. for the same reason, the undercarriage has an extended length of 160cm.
The shape of one skid of the undercarriage is as shown below:

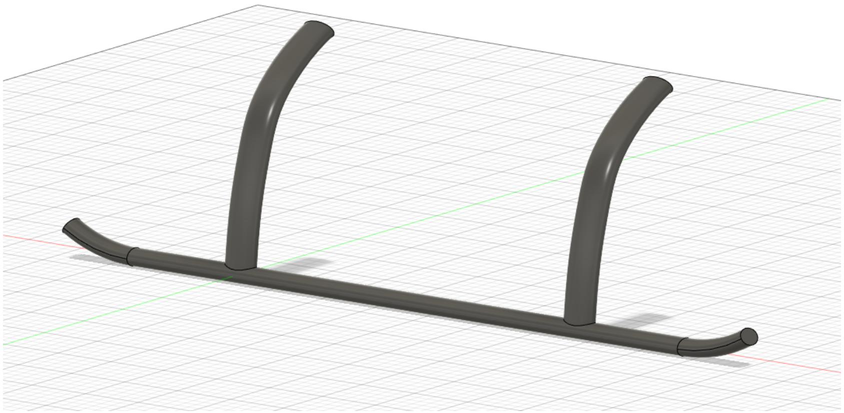


Figure 3. Undercarriage Skid Design

3.3 Composite Layup

90°
45°
−45°
0°
−45°
45°
90°

$\longrightarrow$ $(90/45/-45/0/-45/45/90)$

Utilizing a $(90/45/-45/0/-45/45/90)_s$ layup of unidirectional (UD) fiber tape for the undercarriage of a disaster relief UAV provides numerous critical advantages. This configuration enhances structural strength and stiffness by distributing loads across multiple directions, which is essential for dynamic forces during landings. The multi-directional reinforcement effectively manages complex stresses, offering excellent resistance to torsional, lateral, and out-of-plane forces.
The alternating fiber orientations resist the initiation and propagation of cracks by acting as barriers to crack growth, preventing small damages from escalating into major failures [2, 3]. This reduces wear and tear, minimizing progressive damage and extending the structure's lifespan. The layup also combats delamination, with multiple fiber interfaces increasing inter-laminar shear strength, thereby maintaining structural integrity over time.

Additionally, the layup maximizes mechanical properties while minimizing weight, enhancing payload capacity and flight endurance. The balanced fiber distribution reduces localized stresses, improving fatigue resistance and ensuring durability in repeated use under varied environmental conditions. The layup's resistance to environmental factors like moisture and temperature variations is crucial for UAVs in harsh disaster relief scenarios.
The design simplifies manufacturing, allowing for efficient production and customization to meet mission-specific requirements. This makes the $(90/45/-45/0/-45/45/90)_s$ layup an ideal choice for disaster relief UAVs, ensuring reliability, efficiency, and longevity in challenging operational environments.

3.4 Attachment to Fuselage

The way the undercarriage is attached to the fuselage is determined through several steps. Ideally the attachment should be on the heavy frames of the fuselage so that the structure is at its strongest, however, if the two arms are connected to the first and last heavy frame,the aft attachment point will go beyond the uniform circular region of the fuselage and step into the contracting tail cone, which will make the undercarriage asymmetric thus losing the advantage of easy to maintain. If the u/c is connected to one of the heavy frames in the middle, the bending moment at the point connecting the arm and the skid will be higher leading to a stronger connection hence higher mass. moreover, if the undercarriage is to be connected to a heavy frame, the connection mechanism will be complicated and increases the complexity of assembling, which is a large drawback in the context of a disaster relief UAV.
Another potential option is to connect the skids to the skin using simple bolts and screws, which are easy to work with and assemble. This approach requires finite element analysis (FEA) to justify its feasibility. With this in mind, an FEA was conducted in Ansys Mechanical The simulation conditions involved using a section of the fuselage that is 3 cm wider than the attachment plate to represent the fuselage structure. The front and back faces of this section were fixed, and the screw holes in the undercarriage attachment plate were also fixed. According to the landing conditions, a load equal to 1.6 times the maximum takeoff weight, which is 2511 N, was applied at the tip of the skid at a 7-degree angle with respect to both the XY plane and the YZ plane. The material is considered to be uniform in the FEA for the undercarriage because the $(90/45/-45/0/-45/45/90)_s$ composite layup provides a balanced and symmetrical configuration, ensuring that the composite exhibits consistent mechanical behavior across its plane. By averaging the directional properties of the different fiber orientations, the composite can be modeled with uniform material properties in FEA, simplifying the analysis while maintaining accuracy. This approach allows for the effective prediction of the undercarriage's response to complex loading conditions. After the first simulation, some iterations of reduced mesh size is done as mesh convergence studies. The final converged minimum safety factor based on stress is 4.046, which is in the acceptable range, indicating the design valid.
The results of the FEA is shown as below, and the point with red lable is the point with minimum safety factor:

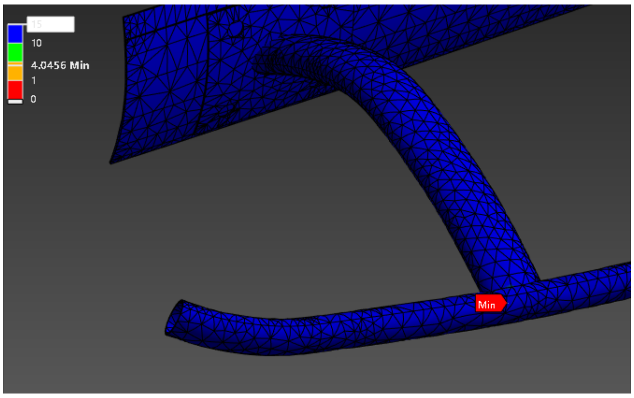


Figure 4. Undercarriage Attachment Study

3.5 FEA analysis

when analysing the effectiveness of the design and how many repetitions are required, another FAE analysis is performed. while the focus of the first FEA was on the attachment and region near the fuselage, this analysis takes a closer look at the skid it self. with the loading conditions staying the same, the boundary condition changed: only the screw holes were chosen to be fixed. Multiple simulations with different mesh size was performed, and the resulting safety factor is 2.5 at the attachment plate screw hole. this is low but within acceptable range and in actual operations the skin of the fuselage and the bolt will act as strengthening mechanism to prevent potential failure. The final converged result is shown below:

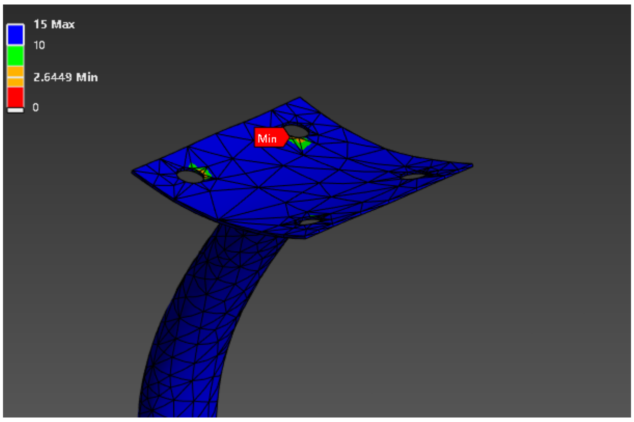


Figure 5. Final Undercarriage Design

3.6 Final Design

With all the above analysis, the final design of the undercarriage is as shown below:

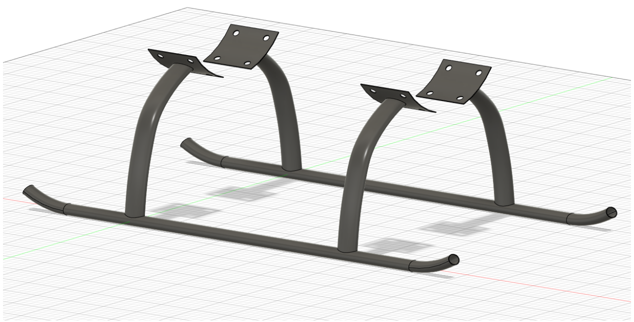


Figure 6. Final Undercarriage Design

4 Boom Loads

2.45 m long booms are located at 1.15 m from the fuselage centreline[4]. The boom is idealised as a front and rear cantilever beam as the wing acts as a fixed end. The front boom extends1.05 m outward from wing flexural axis and has a rotor attached at the free end [4, 5, 6]. The rear boom extends 1.4 m rearward from the flexural axis and has a rotor attached at 1.05 m [4]. The rear boom also has reaction forces at the free end due to tail loads.
During cruise, only boom and rotor inertia are considered along with the tail loads, which are computed in [7]. During takeoff and landing, it is assumed that each rotor generates $L = W/4$, which is then scaled by the relevant load factor [7]. In the OEI case, each of the three remaining rotors must generate $L = W/3$, scaled by load factors, for sufficient lift. Boom drag during VTOL is negligible[8]. Meanwhile, each rotor is assumed

to weigh 3 kg and the boom weighs 2 kg [6]. Inertia is split between the front and rear boom by considering length ratios. Load distributions and internal loads can now be found. As rotor loads are modelled as point loads and inertia modelled as uniform distributed loads, literature results for cantilever beam internal loads are used and superposed to obtain Fig. 7, 8, 9 and 10. Note that the front boom generates no torque and rear boom generates maximum 100 N and minimum -50 N of torque. Also note that the shear forces for front and rear boom are defined in same direction, while bending moment is defined in opposing directions. Only the most constraining cases for takeoff, cruise and landing are plotted. Since drag is neglected, takeoff and landing track the same line.

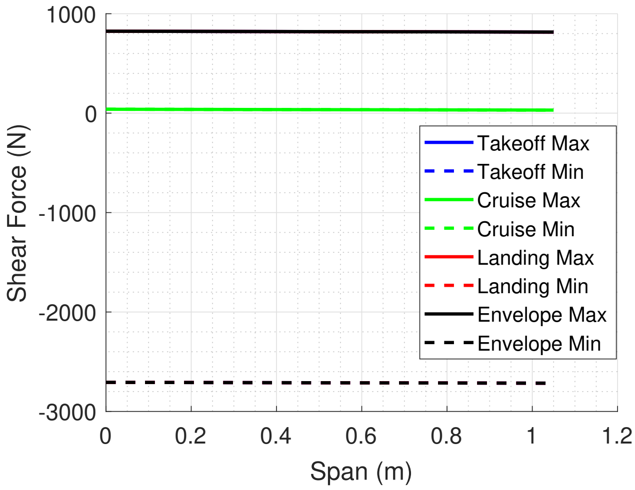


Figure 7. Front boom shear force

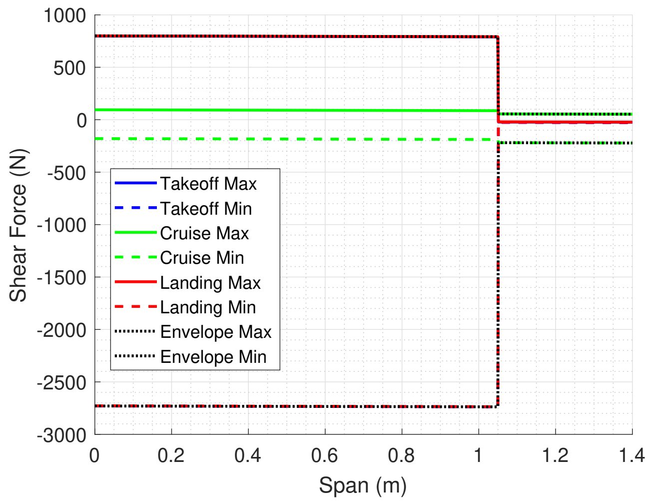


Figure 8. Rear boom shear force

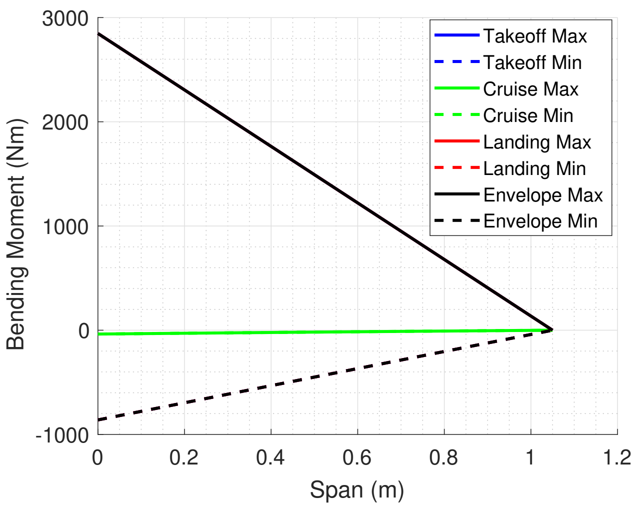


Figure 9. Front boom bending moment

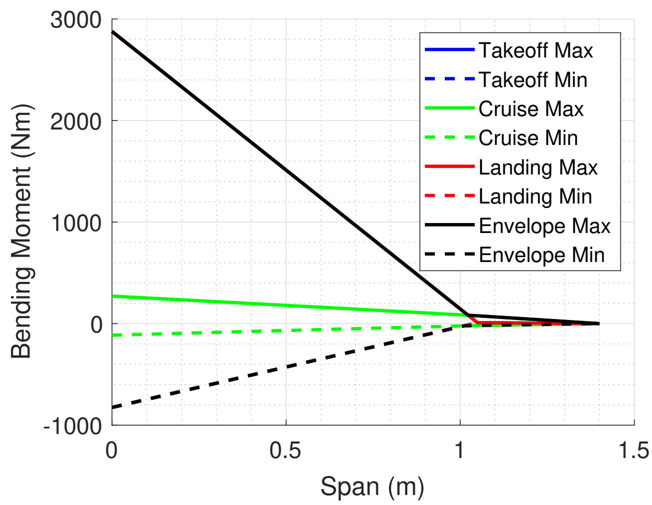


Figure 10. Rear boom bending moment

5 Dynamic Analysis

The natural frequency of a UAV (Unmanned Aerial Vehicle) plays a critical role in its overall performance and operational stability. Natural frequency is the rate at which a system oscillates when not subjected to a continuous or external force, and it is inherently linked to the structural characteristics of the UAV. If the UAV's operational frequencies, including those from engine vibrations, aerodynamic forces, or environmental disturbances, coincide with its natural frequencies, resonance can occur. Resonance can lead to excessive vibrations that compromise the structural integrity of the UAV, potentially causing fatigue, failure of components, or loss of control. Therefore, it is essential to design UAVs with natural frequencies that are well-separated from any operational or environmental frequencies to ensure that resonance and its detrimental effects are avoided.

5.1 Simplified Model

For the UAV configuration depicted, where the booms extend from the fuselage to support the wings and propellers, simplifying these booms as cantilever beams and using their natural frequency to represent the UAV's overall natural frequency is a justified approach. The booms are the primary structural elements that experience significant aerodynamic, thrust, and gravitational loads, functioning similarly to cantilever beams fixed at the fuselage and free at the ends. This simplification is practical because the booms, being long and slender, are likely the most flexible components, significantly influencing the UAV's vibration characteristics. By modeling the booms as cantilever beams, we capture the primary load paths and vibration modes, thus effectively reducing the complex dynamics of the UAV into a more manageable analytical form.
Considering the booms' natural frequency as a representative metric for the entire UAV ensures that we address the most significant structural elements that dictate the UAV's vibration behavior. This conservative approach is not only practical but also essential for ensuring the structural integrity and operational stability of the UAV. By avoiding resonance with operational frequencies, such as those from the propellers or external disturbances, we enhance the UAV's reliability and extend its service life. This method should provide a balance between simplicity and accuracy.

Original Configuration

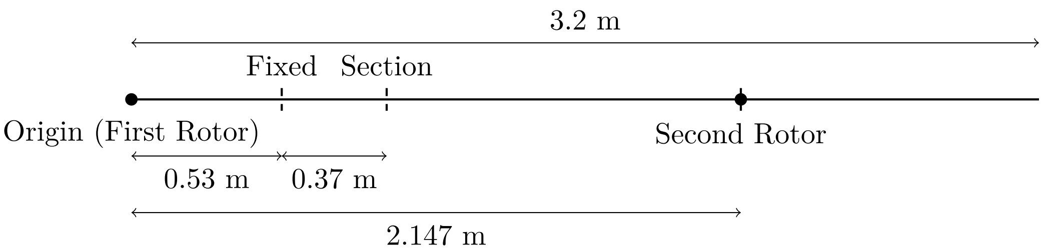


Figure 11. Original configuration of the UAV's boom with dimensions and rotor positions.

The natural frequency is calculated as below[9]

Natural Frequncy Formula

The natural frequency f_n is calculated using:

$$f_n = \frac{K_n}{2\pi}\sqrt{\frac{EIg}{wL^4}}$$

- Mode vibration constants K_n:

$$K_1 = 3.51601527$$
$$K_2 = 22.0344915$$
$$K_3 = 61.6972144$$
$$K_4 = 120.901916$$
$$K_5 = 200$$

- Young's modulus $E = 175 \times 10^9\,\mathrm{Pa}$
- Density $\rho = 1562\,\mathrm{kg/m^3}$
- Outer diameter $D_{\text{outer}} = 0.076\,\mathrm{m}$
- Inner diameter $D_{\text{inner}} = 0.06\,\mathrm{m}$
- Lengths of segments: $L_1 = 0.53\,\mathrm{m}$ and $L_2 = 1.247\,\mathrm{m}$
- Gravitational acceleration $g = 9.81\,\mathrm{m/s^2}$

The weight per unit length w is:

$$A = \frac{\pi}{4}\left(D_{\text{outer}}^2 - D_{\text{inner}}^2\right)$$
$$A = \frac{\pi}{4}\left(0.076^2 - 0.06^2\right) \approx 1.377 \times 10^{-3}\,\mathrm{m^2}$$
$$w = \rho A g = 1562 \times 1.377 \times 10^{-3} \times 9.81 \approx 21.09\,\mathrm{N/m}$$

The moment of inertia I is:

$$I = \frac{\pi}{64}\left(D_{\text{outer}}^4 - D_{\text{inner}}^4\right)$$

Substituting the values:

$$I = \frac{\pi}{64}\left(0.076^4 - 0.06^4\right) \approx 5.04 \times 10^{-6}\,\mathrm{m^4}$$

Hence for each segment:

Natural Frequencies for Segment $L_1 = 0.53\,\mathrm{m}$

$$f_n \approx \frac{K_n}{2\pi} \times 82.08$$

Mode	Natural Frequency (Hz)
1	$f_{1,1} \approx 45.99$
2	$f_{1,2} \approx 288.16$
3	$f_{1,3} \approx 806.72$
4	$f_{1,4} \approx 1582.72$
5	$f_{1,5} \approx 2612.76$

Table 3. Natural frequencies for the first segment L_1

Natural Frequencies for Segment $L_2 = 1.247\,\mathrm{m}$

$$f_n \approx \frac{K_n}{2\pi} \times 15.04$$

Mode	Natural Frequency (Hz)
1	$f_{2,1} \approx 8.42$
2	$f_{2,2} \approx 52.65$
3	$f_{2,3} \approx 147.38$
4	$f_{2,4} \approx 288.60$
5	$f_{2,5} \approx 476.94$

Table 4. Natural frequencies for the second segment L_2

5.2 FEA Dynamic Analysis

Due to computational limitations, simulating the exact build of aircraft is both time and computational power consuming, thus components that does not play a significant role in the dynamic response can be neglected. hence only the booms, the wing, the fuselage, the undercarriage and the tail will be considered, while the rotors and the propellers will be neglected. the rotor mounts will be simplified as point masses. Due to limitations of the software and student edition, the mesh cannot be refined to a satisfying size and winglets cannot be included. The model used in the simmulation is as shown below

Figure 12. Curing Process

It can be seen that the natural frequencies does not match the simplified model, meaning the initial model was over simplified. From literature, the first few natural frequencies are less important [10], hence the 7th to 12th natural frequencies should be considered. which is: 208 270 304 304 503 1203, these are frequencies to be avoid when choosing the rotor and propellers. This simulation gives important insight to what the frequencies of the rotors should be avoided. Improvements to this study can be done via finding ways to include the winglets in the simulations, further improvements in the future can take aerodynamic and environmental sourced vibrations into account.

6 Manufacturing Process

6.1 Toray 3960-19 Prepreg System

The process of manufacturing aircraft components such as booms, wings, fuselage, undercarriage, tail, and rotor blades using automatic layup machines involves several precise and automated steps.
First, for the wings, tail plane and fuselage, large sheets of composite material are automatically laid onto molds by the layup machine, ensuring uniformity and reducing manual error. The machine precisely places each layer of prepreg according to the design specifications. For components like the booms and undercarriage skids, the layup machine uses specialized molds to shape and layer the materials, wrapping the molds and creating the desired shapes, giving strong, lightweight structures. For rotor blades, the machine meticulously applies the composite layers to create aerodynamic and structurally sound blades that can withstand the stresses of flight. Throughout the process, sensors and software ensure accuracy, detecting any inconsistencies and making real-time adjustments. Although very high in cost, the automatic layup machines significantly enhance the

efficiency and precision of aircraft component manufacturing, resulting in high-quality, reliable parts that meet the high standards. After the layup is done, the half finished components will be sent into an autoclave for curing. Initially, the part undergoes a vacuum application to remove any trapped air and ensure a uniform environment for the curing process. the bagging process is illustrated below [11]:

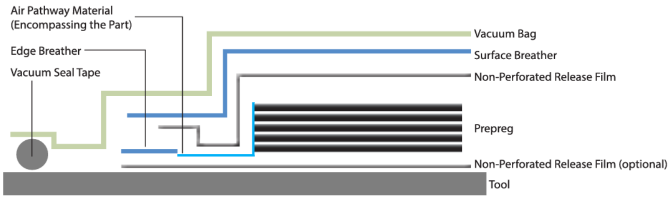


Figure 13. Bagging Process

Once the vacuum is applied, the autoclave pressure is set to 85 psi with a tolerance of +15/-0 psi, corresponding to approximately 586 kPa, creating a controlled atmospheric condition inside the autoclave. As the autoclave pressure reaches 20 psi (138 kPa), the vacuum bag is vented, indicating the start of the curing phase. The temperature inside the autoclave is then ramped up gradually to 355 ± 10°F (180 ± 5°C) at a rate between 1.7 to 4.7°F (1.7 ± 1.1°C) per minute, ensuring a gentle increase to prevent thermal shock. The part is held at this temperature for a duration between 120 to 180 minutes to allow the resin to cure fully. After the curing period, the vessel is cooled down to 140°F (60°C) or lower at a maximum rate of 5°F (2.8°C) per minute before the autoclave pressure is released. This controlled cooling prevents thermal stress and ensures the integrity of the cured composite part. Precise temperature and pressure control with gradual heating and cooling achieves optimal material properties in the final composite product.
The curing process is as shown in the figure below [11]:

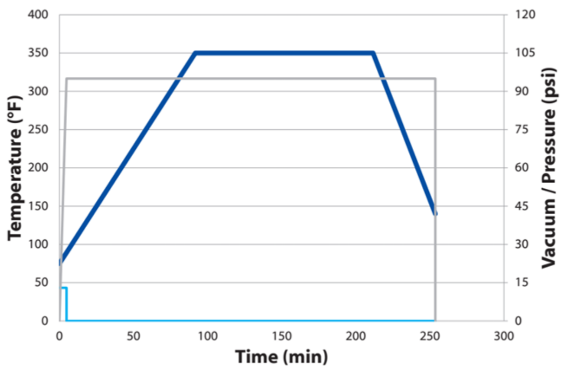


Figure 14. Curing Process

6.2 Al 2024-T4

The manufacturing method for heavy frames are chosen to be CNC (Computer Numerical Control). the heavy frames bear the majority of the load from the payload, and it is a component the UAV cannot risk damaging or malfunctioning, thus CNC is pivotal due to its precision, efficiency, and versatility. In this process, pre-programmed software directs the movement of machinery and tools to cut and shape aluminum alloy into complex, custom-designed frames with high accuracy. The use of CNC ensures consistent quality and allows for intricate designs that would be difficult or impossible to achieve with manual methods. It also minimizes material waste and reduces production time by automating repetitive tasks and complex cutting operations. By employing CNC technology, manufacturers can produce robust, lightweight aluminum frames that meet exact specifications, ensuring structural integrity and optimal performance in various applications.

References

[1] Shivani Chauhan. Halo reach: Fuselage and boom structural design. GDP report 02096006, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[2] Sonali Gyan. Damage-tolerant design optimization of laminated composite structures using dispersion of ply angles by genetic algorithm. 31(12), 2012.

[3] Larry B. Lessard. Damage tolerance of laminated composites containing an open hole and subjected to compressive loadings: Part ii—experiment. 25(1), 1991.

[4] Ruairidh Scott-Brown. Halo reach: Empennage and lateral control design. GDP report 01734080, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[5] Stephen Liu. Halo reach: Wing loads and structural design. GDP report 02015258, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[6] Tharm Chursuk. Halo reach: Optimising weight and balance through comprehensive powerplant sizing for vtol fixed-wing uav. GDP report 02048996, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[7] Jiaxun Leng. Halo reach: V-n diagram, tail, rotor and propeller loads and structural design. GDP report 02015060, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[8] Max F Lawlor. Halo reach: Drag analysis methodology and validity. GDP report 02026199, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[9] Onscale.co. Validation case: Modal analysis of a cantilever beam, 2024. URL `https://onscale.com/help/solve/validation/validation-case-modal-analysis-cantilever-beam/`.

[10] Wei Wenfei. Modal analysis of four rotor aircraft frame. 38(2), 2017.

[11]

Appendix

A Compressiblity effects on an aerofoil at different Mach numbers

Even at subsonic speeds supersonic velocity can be achieved at the suction part of the horizontal stabiliser leading to shocks and possible separation seen in Fig. **??**. This is to be avoided in both the wing and tail for its implication on stall.

B Three Views

C Tail plane moment arm derivation

The equation of the tailplane moment arm, Eq. 10, was derived from the designers interest to minimise wetted area, and therefore drag.

$$S_{wet_h} = \frac{0.95\overline{V_h}\bar{c}S}{l_h}(1.977 + 0.52\frac{t}{c}\big|_h) \tag{10}$$

$$S_{wet_v} = \frac{0.95\overline{V_v}\bar{c}S}{l_v}(1.977 + 0.52\frac{t}{c}\big|_v) \tag{11}$$

where l_v was taken to be approximately $l_h - 0.5$ in metres, and therefore differentiable by l_h, and $\frac{0.95\overline{V_i}\bar{c}S}{l_i}$ is the exposed area of the lifting surfaces.

$$S_{wet_f} = \pi D_f(l_h - \frac{16}{90}L_f) + \pi\frac{16}{90}L_f(R_1 + R_2) \tag{12}$$

where $R_1 = D_f/2$ m and $R_2 = 0.3048$ m, which represent the maximum fuselage radius and the end of fuselage radius. The $\frac{16}{90}L_f$ of the fuselage represents the part of the fuselage which is a frustum.

$$\frac{d\sum S_{wet_i}}{dl_h} = 8.6180 - \frac{541.8613}{l_h^2} - \frac{47.7755}{l_v^2} = 0 \tag{13}$$

The assumption of this derivation is that all the lift surfaces are rectangular as seen in Fig. **??**. This ideas was discarded since the assumptions where not valid for our aircraft and yielded a far back position of the wing with respect to the tail unless the empennage area was increased to $S_h/S \geq 0.35$ and $S_v/S \geq 0.25$ which was deemed unrealistically big.